

2.22

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Exercise 2.22: Tennessee STAR Experiment Data Loading

```
library(PoEdata)
url <- "http://www.principlesofeconometrics.com/poe5/data/rdata/star5_small.rdata"
dest <- file.path(tempdir(), "star5_small.rdata")
download.file(url, dest, mode = "wb")
load(dest)
```

Part (a)

```
data_a <- star5_small$regular == 1 | star5_small$small == 1

df <- star5_small[data_a, ]
mod1 <- lm(totalscore ~ small, data = df)
summary(mod1)
```

```
##
## Call:
## lm(formula = totalscore ~ small, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -189.44  -50.44   -9.44   43.06  324.38
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.675 249.396  <2e-16 ***
## small        12.175       5.369   2.268  0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 74.59 on 773 degrees of freedom
## Multiple R-squared:  0.006608,    Adjusted R-squared:  0.005323
## F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

```

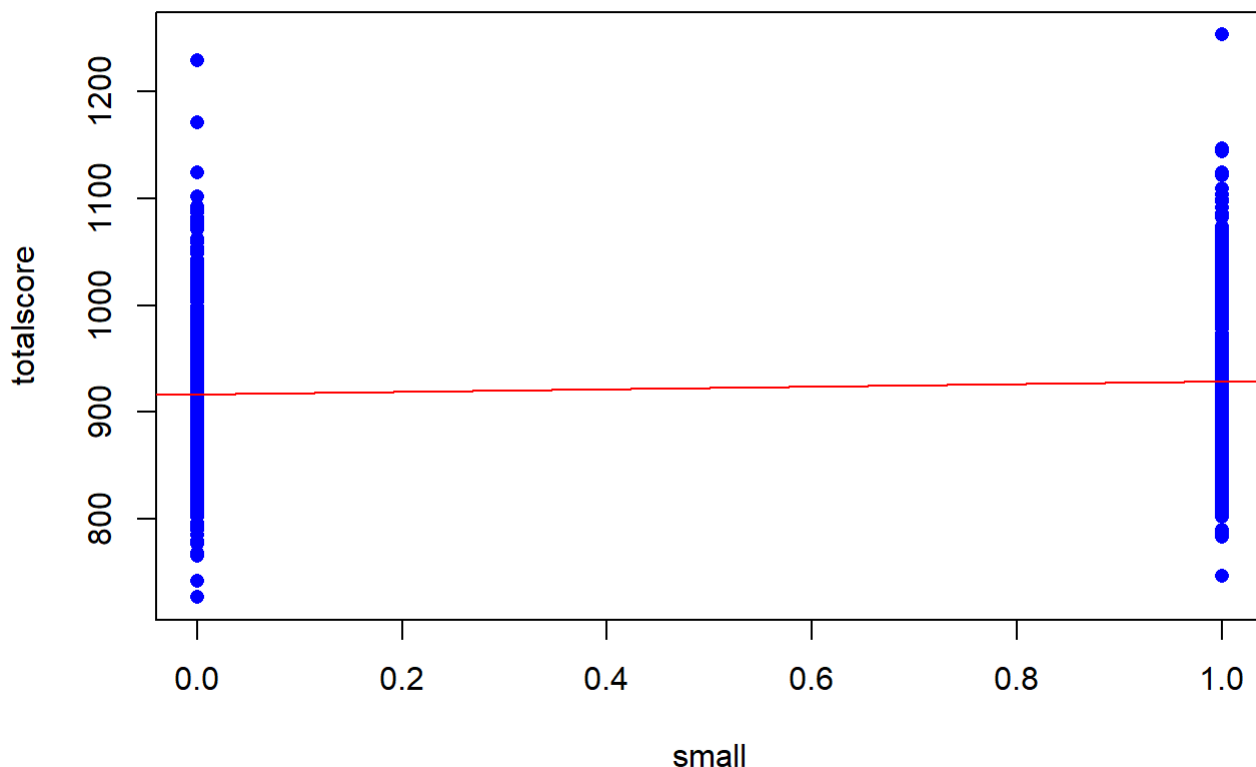
#The average total score representing students in a "regular class"
intercept <- coef(mod1)[1]

#The average difference in total score between "small class" and "regular class".
#If beta2 is positive and significant, it indicates that students in smaller classes perform better.
slope <- coef(mod1)[2]

#Graph
plot(df$small, df$totalscore,
     xlab="small",
     ylab="totalscore",
     main = "Observations and fitted line",
     pch = 16, col = 'blue',
     type = "p")
abline(intercept, slope ,col = 'red')

```

Observations and fitted line



```

cat(
  "b1 is: ", intercept,"when small=0, E[Totalscore|small=0]=b1", "\n",
  "b2 is: ", slope,"when small increase 1,totalscore increase b2", "\n",
  "Since b2>0, small class has higher totalscore, but the points in the two groups are widely dispersed.", "\n",
  "thus this may not Statistically significant", "\n",
  sep = ""
)

```

```
## b1 is: 916.4417,when small=0, E[Totalscore|small=0]=b1
## b2 is: 12.17533,when small increase 1,totalscore increase b2
## Since b2>0, small class has higher totalscore, but the points in the two groups are widely
dispersed.
## thus this may not Statistically significant
```

Part (b)

```
##b.
#read
mod2 <- lm(readscore ~ small, data = df)
summary(mod2)
```

```
##
## Call:
## lm(formula = readscore ~ small, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -74.665 -21.590  -2.665  16.410 194.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  432.665      1.488 290.760 < 2e-16 ***
## small         6.924       2.174   3.185 0.00151 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30.2 on 773 degrees of freedom
## Multiple R-squared:  0.01295,    Adjusted R-squared:  0.01167
## F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
intercept_2 <- coef(mod2)[1]
slope_2      <- coef(mod2)[2]

#math
mod3 <- lm(mathscore ~ small, data = df)
summary(mod3)
```

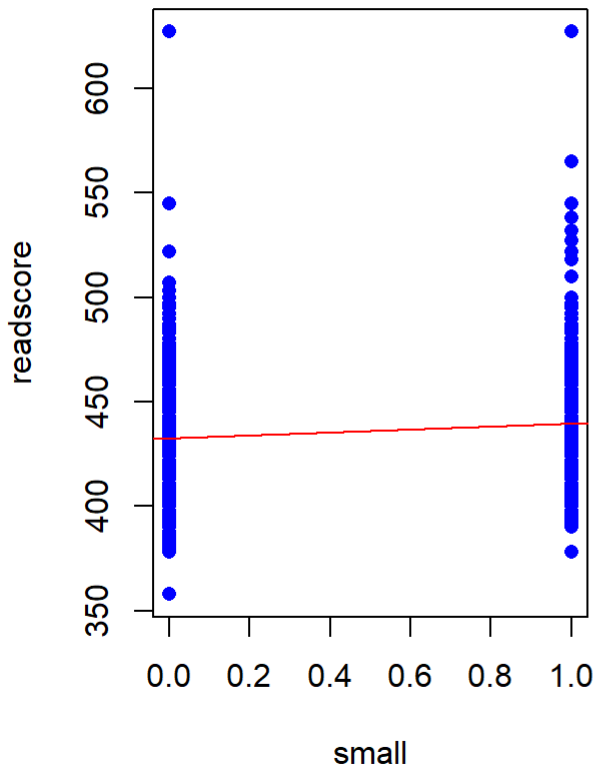
```
##
## Call:
## lm(formula = mathscore ~ small, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -144.777  -35.028   -5.028   29.223  142.223
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.449  197.545  <2e-16 ***
## small        5.251       3.578   1.467   0.143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.71 on 773 degrees of freedom
## Multiple R-squared:  0.002778,    Adjusted R-squared:  0.001488
## F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

```
intercept_3 <- coef(mod3)[1]
slope_3      <- coef(mod3)[2]

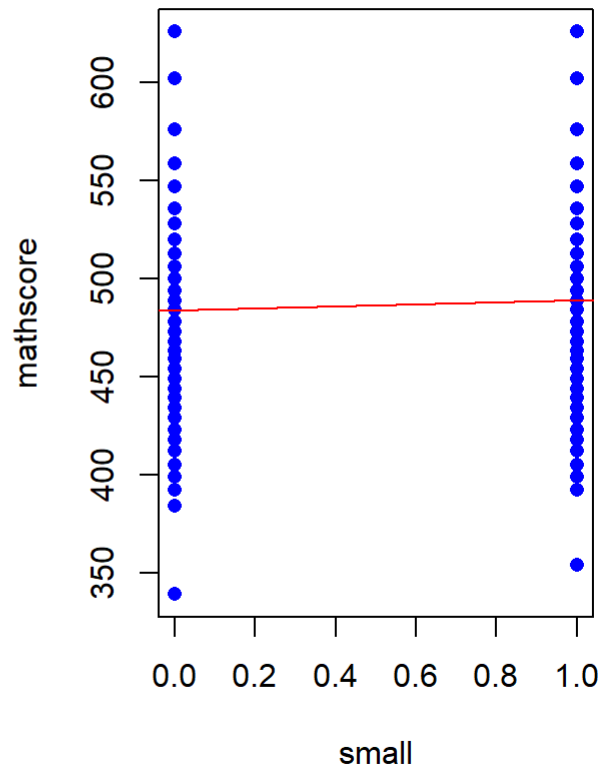
#Graph
par(mfrow = c(1, 2))
plot(df$small, df$readscore,
     xlab="small",
     ylab="readscore",
     main = "Readscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_2,slope_2 ,col = 'red')

plot(df$small, df$mathscore,
     xlab="small",
     ylab="mathscore",
     main = "Mathscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_3,slope_3 ,col = 'red')
```

Readscore



Mathscore



```
par(mfrow = c(1, 1))

cat(
  "b1_read is: ", intercept_2, ". b1_math is: ", intercept_3, "\n",
  "b2_read is: ", slope_2, ". b2_math is: ", slope_3, "\n",
  "Readscore is more improved than Mathscore when class is small", "\n",
  "But it may not Statistically significant", "\n",
  sep = ""
)
```

```
## b1_read is: 432.665. b1_math is: 483.7767
## b2_read is: 6.924483. b2_math is: 5.250849
## Readscore is more improved than Mathscore when class is small
## But it may not Statistically significant
```

Part (c)

```
##c.
idx2 <- star5_small$regular == 1 | star5_small$aide == 1

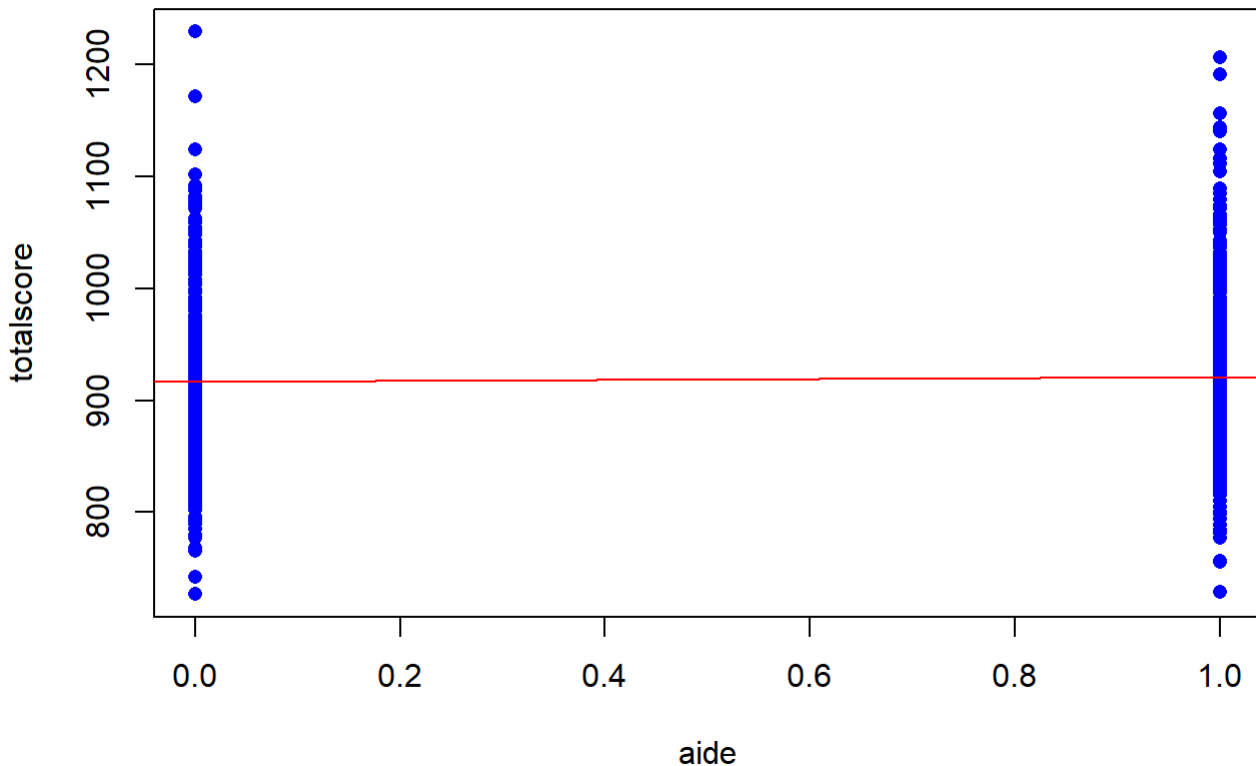
df2 <- star5_small[idx2, ]
mod4 <- lm(totalscore ~ aide, data = df2)
summary(mod4)
```

```
##
## Call:
## lm(formula = totalscore ~ aide, data = df2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -191.748  -50.442   -5.442   40.558  312.558
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.559  257.529  <2e-16 ***
## aide         4.306       4.994   0.862   0.389
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 72.23 on 835 degrees of freedom
## Multiple R-squared:  0.0008898, Adjusted R-squared:  -0.0003068
## F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888
```

```
intercept_4 <- coef(mod4)[1]
slope_4      <- coef(mod4)[2]

#Graph
plot(df2$aide, df2$totalscore,
     xlab="aide",
     ylab="totalscore",
     main = "Observations and fitted line",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_4,slope_4 ,col = 'red')
```

Observations and fitted line



```
cat(
  "b1 is: ", intercept_4," , when aide=0, E[Totalscore|aide=0]=b1", "\n",
  "b2 is: ", slope_4," , when aide increase 1, totalscore increase b2", "\n",
  "Since b2>0, regular class with aide has higher totalscore, but the points in the two group",
  "s are widely dispersed.", "\n",
  "Thus this may not Statistically significant", "\n",
  sep = ""
)
```

```
## b1 is: 916.4417, when aide=0, E[Totalscore|aide=0]=b1
## b2 is: 4.306488, when aide increase 1, totalscore increase b2
## Since b2>0, regular class with aide has higher totalscore, but the points in the two group",
## s are widely dispersed.
## Thus this may not Statistically significant
```

Part (d)

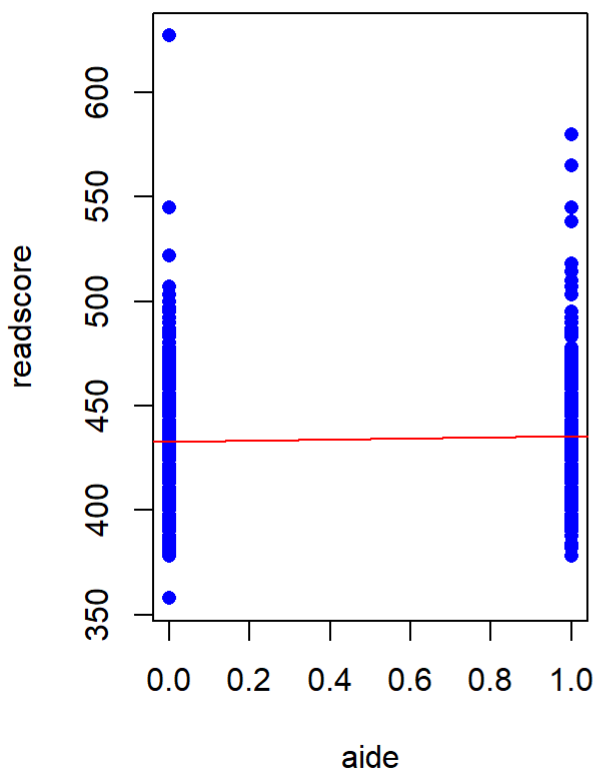
```
##d.
mod5 <- lm(readscore ~ aide, data = df2)
intercept_5 <- coef(mod5)[1]
slope_5      <- coef(mod5)[2]

mod6 <- lm(mathscore ~ aide, data = df2)
intercept_6 <- coef(mod6)[1]
slope_6     <- coef(mod6)[2]

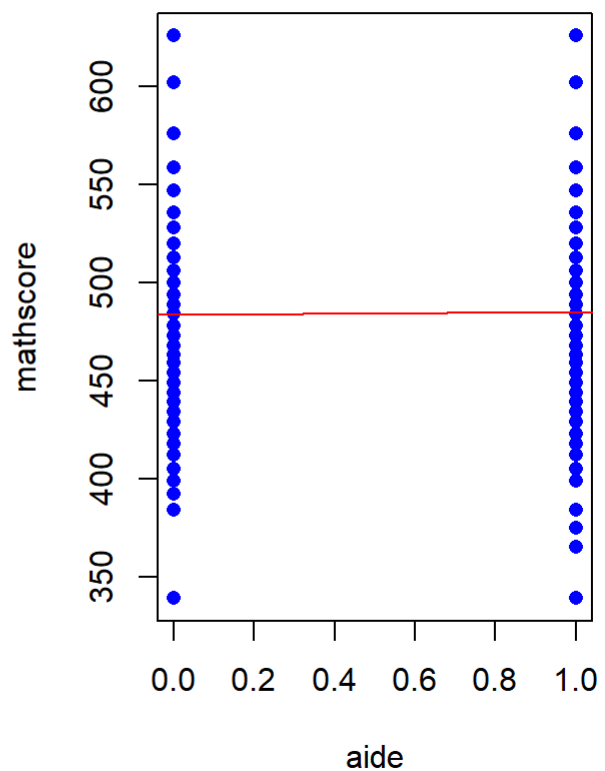
#####
par(mfrow = c(1, 2))
plot(df2$aide, df2$readscore,
     xlab="aide",
     ylab="readscore",
     main = "Readscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_5,slope_5 ,col = 'red')

plot(df2$aide, df2$mathscore,
     xlab="aide",
     ylab="mathscore",
     main = "Mathscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_6,slope_6 ,col = 'red')
```

Readscore



Mathscore




```
par(mfrow = c(1, 1))
```

```
cat(  
  "b1_read is: ", intercept_5, ". b1_math is:", intercept_6, "\n",  
  "b2_read is: ", slope_5, ". b2_math is:", slope_6, "\n",  
  "Readscore is more improved than mathscore when class has aide", "\n",  
  "But it may not Statistically significant", "\n",  
  sep = ""  
)
```

```
## b1_read is: 432.665. b1_math is:483.7767  
## b2_read is: 2.871422. b2_math is:1.435066  
## Readscore is more improved than mathscore when class has aide  
## But it may not Statistically significant
```